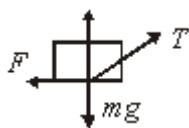


Mark schemes

Q1.

(a)



Correct diagram

B1

1

(b) $40 \cos 30^\circ - F = 25 \times 0.1$

$F = 40 \cos 30^\circ - 2.5 = 32.1 \text{ N}$

Three term equation of motion

M1

Correct equation

A1

AG; correct force from correct working

A1

3

(c) $R + 40 \sin 30^\circ = 25 \times 9.8$

$R = 225 \text{ N}$

Resolving vertically

M1

Correct equation

A1

AG; correct force from correct working

A1

3

(d) $32.1 = 225\mu$

use of $F = \mu R$

M1

$\mu = \frac{32.1}{225} = 0.143$

Correct μ

A1

2

Q2.

(a) $F = 0.8 \times 8 \times 9.8 = 62.72 \text{ N}$

Use of $F = \mu R$ with $R = 8 \times 9.8$

M1

Correct F

A1

2

(b) $7g - T = 7a$

Three term equation of motion for one particle

M1

$T - 62.72 = 8a$

Correct equation

A1

$7g - 62.72 = 15a$

Three term equation of motion for other particle

M1

$a = \frac{7g - 62.72}{15} = 0.392$

Correct equation

A1

AG; correct a from correct working

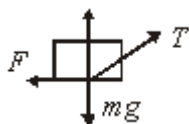
A1

5

[7]

Q3.

(a)



Correct diagram

B1

1

- (b) $40 \cos 30^\circ - F = 25 \times 0.1$
 $F = 40 \cos 30^\circ - 2.5 = 32.1 \text{ N}$
Three term equation of motion

M1

Correct equation

A1

AG; correct force from correct working

A1

3

- (c) $R + 40 \sin 30^\circ = 25 \times 9.8$
 $R = 225 \text{ N}$
Resolving vertically

M1

Correct equation

A1

AG; correct force from correct working

A1

3

- (d) $32.1 = 225\mu$
use of $F = \mu R$

M1

$\mu = \frac{32.1}{225} = 0.143$

Correct μ Exam Papers Practice. All rights reserved.

A1

2

- (e) Friction will decrease as normal reaction
Decrease in friction

B1

decreases

Normal reaction decreases

B1

2

Q4.

- (a) Light or smooth

Acceptable assumption

B1

1

- (b) $5g - T = 5a$

Three term equation of motion for one particle

M1

$$T - 2g = 2a$$

Correct equation

A1

$$3g = 7a$$

Three term equation of motion for other particle

M1

$$a = \frac{3g}{7} = 4.2 \text{ ms}^{-2}$$

Correct equation

A1

AG; correct acceleration from correct working

A1

5

- (c) $T = 2 \times 4.2 + 2 \times 9.8 = 28 \text{ N}$

Substitute $a = 4.2$ into one equation of motion

M1

Correct tension

A1

2

[8]

Q5.

- (a) $200 \sin 30^\circ = T \sin 45^\circ$

$$T = \frac{200 \sin 30^\circ}{\sin 45^\circ} = 141 \text{ N}$$

Resolving horizontally

M1

Correct equation

AG; correct T from correct working

A1

A1

3

(b) $200 \cos 30^\circ + 141 \cos 45^\circ + R = 500 \times 9.8$

Resolving vertically with four terms

M1

Correct values

A1

Correct signs

A1

$R = 4630 \text{ N}$

Correct R

A1

4

[7]

Q6.

(a) $9g - T = 9a$

M1: Equation for one particle

A1: Correct equation

M1A1

$T - 5g = 5a$

M1: Equation for other particle

A1: Correct equation

M1A1

$4g = 14a$

$a = \frac{4g}{14} = 2.8 \text{ ms}^{-2}$

A1: Correct a from correct working

AG

A1

5

(b) $T - 5g = 5 \times 2.8$

M1: Substituting acceleration to find T

M1

$$T = 63 \text{ N}$$

A1: Correct tension

A1

2

(c) $s = \left(\frac{1}{2} \times 2.8 \times 0.5^2 \right)$

M1: Constant acceleration equation with $u = 0$ and $a \neq g$ to find s . Allow $\pm$ answers.

A1: Correct equation

M1A1

$$= 0.35 \text{ m}$$

A1: Correct distance

A1

$$\text{Total} = 2 \times 0.35 = 0.7 \text{ m}$$

A1: Doubling their distance to get total distance apart

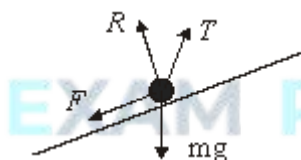
A1ft

4

[11]

Q7.

(a)



B1: Correct force diagram

B1

1

(b) $R + 20 \sin 30^\circ = 6g \cos 10^\circ$

M1: Resolving perpendicular to the slope with 3 terms

A1: Correct equation

M1A1

$$R = 6g \cos 10^\circ - 20 \sin 30^\circ$$

dM1: Solving for R

dM1

$$R = 47.9 \text{ N (to 3 sf)} \quad \text{AG}$$

A1: Correct R from correct working

A1

4

(c) $F = \mu R$

M1: Use of $F = \mu R$

M1

$$6 \times 0.4 = 20 \cos 30^\circ - 6g \sin 10^\circ - \mu R$$

M1: Resolving parallel to slope to get 4 term equation of motion

A1: Correct equation

M1A1

$$\mu R = 4.710$$

A1: Correct $F / \mu R$

A1

$$\mu = \frac{4.710}{47.91} = 0.0983$$

dM1: Solving for μ

dM1

A1: AWR 0.098

A1

6

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[11]

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Q8.

(a) $R = 2 \times 9.8, \quad F = \frac{1}{7} \times 2 \times 9.8 = 2.8 \text{ N}$

CAO M1: full method with 2×9.8 , accept inequality

M1A1

2

(b) (i) $P = 1, \quad F = 1 \text{ N}$

B1

1

(ii) $P = 5, \quad F = 2.8$

Used

B1

$$5 - 2.8 = 2a$$

M1: 3 terms, with forces subtracted

M1A1

$$a = 1.1 \text{ ms}^{-2}$$

ft one slip

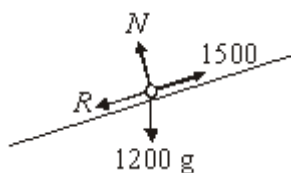
A1F

4

[7]

Q9.

(a) (i)



3 correct & labelled

B1

All correct & labelled, no extras

B1

2

Ignore additional components of weight

(ii) $1500 = R + 1200 \times 9.8 \times \frac{1}{14}$
M1: 3 terms with component attempted
($\alpha = 4.096$) – 1 each error

M1A2

$$R = 660$$

AWRT

A1

4

(iii) 15 ms^{-1}

B1

1

(b) (i) $\frac{1500}{1200}$
Accept $\pm$

M1

Mag of retardation = 1.25 ms^{-2}

A1

2

(ii) $0 = 15 + (-1.25) t$

velocities correct, but accept $\pm$ acceleration

M1

$t = 27.3 \text{ sec} \quad (27.27)$

ft one slip

A1F

2

[11]

Q10.

(a) $5mg - T = 5m \frac{g}{4}$

3 terms, recognisable, accept m missing or g missing once

M1

all correct & algebraic

A1

$T = 15 m \frac{g}{4} = 3.75 mg$

ft 9.8 used, one g missing, m missing, sign error provided T positive

A1F

3

(b) $T - kmg = km \frac{g}{4}$

as in (a)

M1A1

$k = 3$

CAO

following fully correct work in (a) & (b)

A1

3

(c) Force = $2T = \frac{15mg}{2} = 7.5 mg$

B1F

1

(d) $t = \frac{2}{3}$

$x = 0 + \frac{1}{2} \times \frac{g}{4} \times \frac{4}{9}$ accept $\left(\frac{2}{3}\right)^2$

method for x , and $\frac{g}{4}$ used for acceleration

M1

accept 9.8 substituted

A1

$d = 2x = \frac{g}{9}$

Fully correct, in terms of g

A1

3

[10]

Q11.

(a) $5 = 10 \cos \theta^\circ$
Component attempted

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M1

$\theta = 60$

AG

A1

2

(b) $Q = 10 \sin \theta$
Component attempted

M1

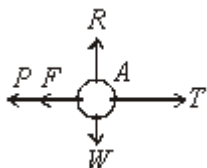
$Q = 5\sqrt{3} \text{ N} = 8.66 \text{ N}$

Accept either form

A1

Q12.

(a) (i)



For any 3 forces correct

B1

*For the rest correct, and no extras
(accept F in either direction, but not μR)*

B1

2

(ii) $R = 0.2 \times g$

$F = 0.5 \times R$

For equations for both R and F

M1

$F = 0.98\text{N}$

AG

A1

2

(iii) $T = 0.3 \times 9.8 = 2.94\text{N}$

M1A1

2

(iv) $P + F = T$

M1

least $P = 1.96\text{N}$

ft T provided $P > 0$

A1F

2

(b) (i) $2.94 - T = 0.3a$

M1A1

$$T - 0.98 = 0.2a$$

M1 either equation

(whole string method: 'correct' equation, M1 A1, answer A1; max 3 / 5)

A1

$$1.96 = 0.5a$$

m1

$$a = 3.92\text{ms}^{-2}$$

ft one error

A1F

5

(ii) $0.1 = \frac{1}{2} \times 3.92 \times t^2$

Use of candidate's acceleration, if 9.8

M1

$$t = 0.226 \text{ (0.22588)}$$

*ft candidate's acceleration (not 9.8) if
a = 1.96 used, t = 0.319*

A1F

2

[15]

Q13.

(a) $\mathbf{F} = \begin{pmatrix} 6 \\ -2.5 \end{pmatrix} \text{ N}$

B1 each component

B1 B1

2

(b) $|\mathbf{F}| = \sqrt{6^2 + (-2.5)^2}$

Must see +

M1

$$= 6.5 \text{ N}$$

ft from vector in (a)

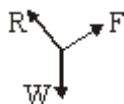
A1F

2

[4]

Q14.

(a)



Three forces labelled & with arrows, W , mg or 35 g vertical (or 2 components of W), R and F perpendicular (ignore pairs of components of existing forces)

B1

1

(b) $R = 35g \times \cos 25^\circ$

Component attempted & g present for M1

M1A1

$R = 311\text{ N}$

($R = 310.86$) accept AWRT 311

A1F

3

(c) $F = 35g \times \sin 25^\circ$

Component attempted & g present for M1 & acceleration zero

M1A1

$(= 144.96)$

$F = \mu R, 144.96 = \mu \times 310.86$

Use of friction law with candidate's values, must have tried to find F

M1

$\mu = 0.466$

ft R and F , provided $\mu > 0$

M1A0 if $F < \mu R$ used

SC accept use of $\mu = \tan \theta$

A1F

4

[8]

Q15.

(a) $1200 - R = 1000 \times 0.25$

M1: all relevant terms used

M1A1

$$R = 950 \text{ N}$$

f.t. one slip if $R > 0$

A1F

3

(b) $2100 - 1000g \times 0.1 - 950 = 1000 \times a$

M1: 3 or 4 terms considered

M1

*−1 each term incorrect (any error),
or missing, $\alpha = 5.74^\circ$*

A1A1

$$a = 0.17 \text{ ms}^{-2}$$

f.t. one error if 4 terms considered

A1F

4

[7]

Q16.

(a) (i) $T = 0.6a$

Either equation (M1 A0 for use of $0.1ga$)

M1

$$0.1g - T = 0.1a$$

SC who;e strng method:

$$0.1g = 0.7a \text{ M1A1 (total mass used)}$$

A1

A1

$$0.1g = 0.7a$$

$$a = 1.4 \text{ A1, max3/5}$$

m1

$$a = 1.4 \text{ ms}^{-2}$$

A1

5

(ii) $T = 0.6 \times 1.4 = 0.84 \text{ N}$

Dependent on M1 gained in (a),

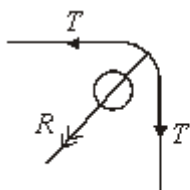
*Or, s.c. can gain M1 (from (a)) A1 here if
equations involving T not found in (a)*

Max M1 A1

A1

1

(iii)



$$R = 2T \cos 45^\circ$$

M1

recognising 2 tensions involved

B1

M1: attempt at Pythagoras or at a component of T

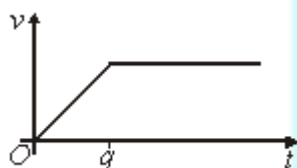
$$= 1.19 \text{ N}$$

A1: f.t. tension

A1F

3

(b)



1st line sloping and through O

B1

2nd line horizontal

B1

label at t q

B1

3

[12]

Q17.

- (a) Sum of vertical forces = zero
used

M1

$$4\cos 60^\circ + 8\cos 60^\circ - q = 0$$

M1 for equation, 3 forces

M1 A1

$$q = 6$$

cao

A1

4

(b) horizontal:

$$R = 3 + 8 \cos 30^\circ - 4 \cos 30^\circ$$

M1A1

$$R = 6.46 \text{ or } 6.47$$

A1F

3

[7]

Q18.

(a) (i) $B: T = mg = 0.1 \times 9.8 = 0.98 \text{ N}$
cao

B1

1

(ii) $A: F = T (= 0.98) \quad R = 0.5 \times 9.8$

B1

$$\mu \times 0.5 \times 9.8 = 0.98$$

Condone inequality

M1

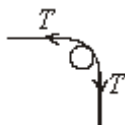
$$\mu = 0.2$$

cao

A1

3

(iii)

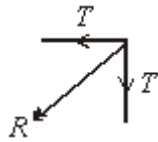


Must label T or 0.98 N

B1

1

(iv)



$$R = \sqrt{T^2 + T^2} \text{ or } 2T \cos 45^\circ$$

M1

$$R = 1.39\text{N}$$

A1

2

- (b) (i) $A: T - 0.98 = 0.5a$
Either equation for M1

M1A1

$$B: 2 \times 0.1 \times 9.8 - T = 0.2a$$

A1

$$0.98 = 0.7a$$

m1

$$a = 1.4$$

cao

A1

5

- (ii) $T = 0.98 + 0.5 \times 1.4, \quad T = 1.68$
ft accuracy error in equation

A1F

1

(iii) $s = ut + \frac{1}{2}at^2$

$$0.7 = 0 + \frac{1}{2} \times 1.4t^2$$

M1

$$t = 1$$

A1

2

[15]

Q19.

(a) (i) $T = mg$

Accept $g = 9.8$ (or 10) used throughout (a)

B1

1

(ii) $F = T (= mg)$

B1

$$R = 2mg$$

B1

$$F \leq \mu R, \quad mg \leq 2mg\mu, \quad \mu \geq \frac{1}{2}$$

*Inequality needed, or convincing verbal explanation,
ft if μ numeric and positive*

B1F

3

(b) $T = 2ma$

M1 either equation

M1A1

$$mg - T = ma$$

Accept $g = 9.8$ or 10 for either A1

A1

$$2mg - 2T = T$$

Full method for T

m1

$$T = \frac{2mg}{3}$$

ft one slip

*If friction included, can either earn M1A1 for Q ,
A0 for P , m1A0.*

SC *Whole string method can earn m1 for $mg = 3ma$
and A1F for reult for T if M1 already earned*

A1F

5

(accept: $T = 6.53m$ if 9.8 used, $T = 6.67m$ if 10 used,

$T = 6.54m$ if 9.8 used and a found first)

[9]

Q20.

(a) $6 = a\cos 60^\circ + 8\cos 60^\circ$

3 terms & attempt at components for M1 correct components

M1

$$6 = \frac{a}{2} + 4$$

signs correct

A1A1

$$a = 4$$

A1

4

(b) $\pm(8\cos 30^\circ - a\cos 30^\circ)$

Components parallel to Oy only for M1 signs correct, 8 and a (or 4) used

M1A1

$$3.46 \text{ or } 3.47$$

ft one error e.g. sign or incorrect a

A1F

3

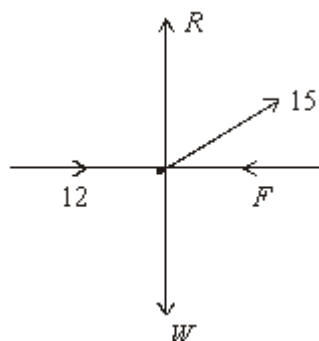
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[7]

Q21.

(a)



any 3 correct, including arrows and labels

B1

all correct, including arrows and labels

B1

2

accept mg for W

*accept either 15 N **or** components, but **not** both*

(b) $F = 12 + 15 \cos 30^\circ$

for M1, both terms on RHS and sin/cos

M1A1

$$= 12 + 12.99$$

$$F = 25 \text{ N}$$

CAO

A1

3

(c) $25 = \frac{1}{3} R$

M1A0 if inequality used

M1

$$R = 75 \text{ N}$$

*AWRT; ft F; accept use of c's F from (b);
accept 0.333 for μ*

A1F

2

(b) $R + 15 \cos 60^\circ = W$

3 terms and attempt at component

$Wg \Rightarrow M1A0A0$

M1

$$75 + 7 \frac{1}{2} = W$$

ft R

A1F

$$W = 82 \frac{1}{2} \text{ N}$$

AWRT 82.5, ft R or one slip

A1F

3

Q22.

- (a) (i) trailer: $\pm S + 100 = 250 \times 0.5$
M1 for 3 terms, A1 all correct, allow $\pm S$
g included with mass $\Rightarrow M1A0$

M1A1

$$S = 25\text{N}$$

ft one error (including g present) provided $S > 0$
If S negative, accept sign changed for answer

A1F

3

- (ii) **car + trailer:**

M1

$$500 + 100 + R = 150 \times 0.5$$

All terms for M1, including appropriate mass
g included with mass $\Rightarrow M1 A0 A1$

or: car:

$$R + 500 - S = 1250 \times 0.5$$

-1 each error; ft S if used

A2, 1, 0

$$R = 150\text{ N}$$

ft one slip, for $R > 0$

If R consistently in wrong direction, need explanatory comment re direction for A1, otherwise M1 A1 A0 A0 ($R < 0$)

A1F

4

- (b) $T = 100\text{ N}$

CAO

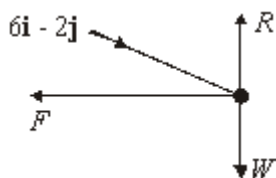
B1

1

[8]

Q23.

- (a)



B1

1

(b) $6 - F = 1.5 \times 0.6$
3 terms for M1

M1 A1

$F = 5.1\text{N}$
CAO

A1

3

(c) $R = 2 + 1.5 \times 9.8$
RHS, 2 forces for M1

M1A1F

$R = 16.7\text{N}$

A1F

3

(d) $F = \mu R$
 $5.1 = \mu \times 16.7$

M1

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$\mu = 0.305$

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A1F

2

[9]

Q24.

(a) $A: \quad 0.21g - T = 0.21a$
'whole string' method 3/5

M1A1

$B: \quad T - 0.14g = 0.14a$
M1A1 for equation, A1 for answer

A1

$$0.07g = 0.35^a$$

M1

$$a = 1.96 \text{ ms}^2$$

CAO

A1

5

(b) $v = 0 + 1.96 \times 2$

M1

$$v = 3.92 \text{ ms}^{-1}$$

A1

2

(c) $0 = 3.92 - 9.8t$

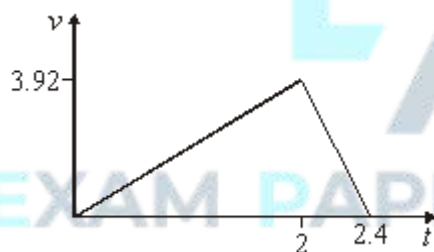
M1

$$t = 0.4 \text{ sec}$$

A1F

2

(d)



Two straight lines

B1

Appropriate gradients

B1

Labelling

B1

3

[12]

Q25.

(a) $150g - T = 150 a$

Three term equation of motion for tank.

M1

$$T - 200g \sin 30^\circ = 200 a$$

Correct equation

A1

$$150g - 200a - 200g \sin 30^\circ = 150 a$$

Three term equation of motion for trolley

M1

Correct equation

A1

$$a = \frac{150g - 200g \sin 30^\circ}{350} = 1.4 \text{ ms}^{-2}$$

Solving for a

m1

Correct equation for a

A1

ag *Correct a from correct working.*

A1

7

(b) $T = mg$

Equations of motion for both bodies.

M1

$$T = 200g \sin 30^\circ$$

Correct equation for tank.

A1

Correct equation for trolley

A1

$$m = 200 \sin 30^\circ = 100 \text{ kg}$$

Correct mass

A1

4

[11]

Q26.

(a) $R = 2 \times 9.8 \cos 40^\circ$

Resolving perpendicular to the slope

M1

$$F = 0.3 \times 2 \times 9.8 \cos 40^\circ$$

Use of $F = \mu R$

M1

$$= 4.50 \text{ N}$$

Correct F

A1

3

(b) $2a = -2 \times 9.8 \sin 40^\circ - 4.50$

Three term equation of motion

M1

$$a = \frac{-2 \times 9.8 \sin 40^\circ - 4.50}{2} = -8.55 \text{ ms}^{-2}$$

Correct acceleration

A1

3

[6]

Q27.

(a) $T - 800 \times 9.8 = 800 \times 0.2$

Three term equation of motion

M1

Correct equation

A1

$$T = 7840 + 160 = 8000 \text{ N}$$

ag *Correct tension*

A1

3

(b) $T = 800 \times 9.8 = 800 \times (0.2)$

Three term equation of motion

M1

$$T = 7840 - 160 = 7680 \text{ N}$$

Correct tension

A1
2

(c) $T = 800 \times 9.8 = 7840 \text{ N}$
Correct tension

B1
1

[6]

Q28.

(a) $R = 20 \times 9.8 = 196 \text{ N}$
Finding R and using $F = \mu R$

M1

$F = 0.15 \times 196 = 29.4 \text{ N}$
Correct friction

A1
2

(b) $T - 29.4 = 20a$
Equation of motion for A

Correct equation

M1

A1

$5 \times 9.8 - T = 5a$
Equation of motion for B

M1

Correct equation

A1

$49 - (20a + 29.4) = 5a$

$a = \frac{19.6}{25} = 0.784 \text{ ms}^{-2}$

ag *Correct acceleration from correct working*

A1
5

(c) $T = 20 \times 0.784 + 29.4 = 45.1 \text{ N}$
Substituting a into one equation of motion

M1

Correct tension

A1

2

(d) $s = \frac{1}{2} \times 0.784 \times 3^2$

Use of constant acceleration equation with $u = 0$

M1

$= 3.53 \text{ m}$

Correct equation

A1

Correct distance

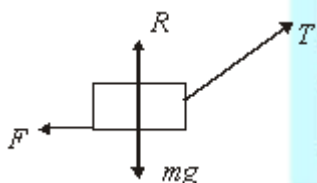
A1

3

[12]

Q29.

(a)



Correct force diagram

B1

1

(b) $R + T \sin 30^\circ = 200 \times 9.8$

Three term equation from resolving vertically

M1

$R + 0.5 T = 1960$

$R = 1960 - 0.5 T$

Correct equation

A1

ag Correct result from correct working

A1

3

(c) $F = T \cos 30^\circ$

Resolving horizontally

M1

$$T \cos 30^\circ = 0.6(1960 - 0.5T)$$

Use of $F = \mu R$

M1

$$T (\cos 30^\circ + 0.3) = 1176$$

Correct equation

A1

Solving for T

m1

$$T = \frac{1176}{(\cos 30^\circ + 0.3)} = 1010 \text{ N}$$

Correct T

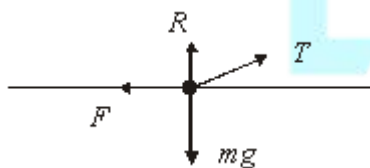
A1

5

[9]

Q30.

(a)



Correct force diagram

B1

1

(b) $R + T \sin 40^\circ = 50 \times 9.8$

$$R = 490 - T \sin 40^\circ$$

Three term equation of motion

M1

Correct equation

A1

Correct expression for R

Use of $F R = \mu$

A1

3

(c) $F = 0.6 (490 - T \sin 40^\circ)$

Use of $F = \mu R$

M1

$= 294 - 0.6 T \sin 40^\circ$

ag Correct result from correct working

A1

2

(d) $T \cos 40^\circ - (294 - 0.6T \sin 40^\circ) = 50 \times 0.5$

$$T = \frac{319}{\cos 40^\circ + 0.6 \sin 40^\circ} = 277 \text{ N}$$

Four term equation of motion

M1

Correct equation

A1

Solving for T

M1

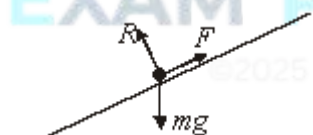
Correct T

A1

4

[10]

Q31.



(a)

B1

1

(b) $R = 5 \times 9.8 \cos 40^\circ = 37.5 \text{ N}$

Correct force diagram

Resolving perpendicular to slope

Correct R

M1 A1

2

(c) $F = 0.2R = 7.51 \text{ N}$

Using $F = \mu R$

Correct F from correct working

M1 A1

2

(d) $5 \times 9.8 \sin 40^\circ - F = 5a$

Resolving parallel to slope to give 3 term equation of motion

M1 A1

Correct equation

$$a = \frac{5 \times 9.8 \sin 40^\circ - F}{5} = 4.80 \text{ ms}^{-2}$$

m1

Solving for a

Correct a from correct working

A1

4

(e) $10^2 = 2^2 + 2 \times 4.80s$

Forming constant acceleration equation

Correct equation

M1 A1

$$s = \frac{100 - 4}{9.6} = 10.0 \text{ m}$$

Correct s

A1

3

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[12]

Q32.

(a) $14a = 14g \sin 45^\circ - T$

Equation of motion for one particle

M1

$$6a = T - 6g$$

Correct equation

A1

$$14a = 14g \sin 45^\circ - (6a + 6g)$$

Equation of motion for other particle

Correct equation

M1 A1

$$a = \frac{14g \sin 45^\circ - 6g}{20} = 1.91 \text{ ms}^{-2}$$

Correct a from correct working

M1 A1

6

(b) $T = mg$

Equation for one particle

M1

$$T = 14g \cos 45^\circ$$

*Equation of other particle
correct m*

M1 A1

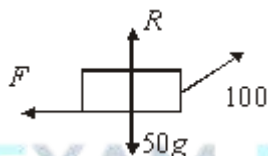
$$m = 14 \cos 45^\circ = 9.90 \text{ kg}$$

A1

4

[10]

Q33.



(a) *Correct force diagram*

B1

1

(b) $50 \times 9.8 = R + 100 \sin 30^\circ$

*Resolving vertically
Correct equation*

M1 A1

$$R = 440 \text{ N}$$

Correct F from correct working

A1

3

(c) $100 \cos 30^\circ \leq \mu \times 440$

Use of $F \leq \mu R$ or $F = \mu R$

M1

$$\therefore k = 0.197$$

A1

$$\mu \geq 0.197$$

Correct equation

Correct k from correct working

A1

(d) $50a = 100 \cos 30^\circ - 0.1 \times 440$

Resolving horizontally to obtain a 3 term equation of motion

M1

3

Correct equation

A1

Solving for a

m1

$$a = 0.852 \text{ ms}^{-2}$$

Correct a

Allow 0.680 or 0.681

A1

4

[11]

Q34.

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(a) (i) $10 \times 9.8 \times 0.5 = 2T$

Moment about pivot with 2 terms

correct revision from correct working

M1

$$T = 24.5$$

Correct moment equation

A1 A1

3

(ii) $10 \times 9.8 \times 0.5 + 40 \times 9.8 \times 3 = 2T$

Moments about pivot with 3 terms

Correct tension from correct working

Correct moment equation

M1 A1

$$T = 613 \text{ (to 3 sf)}$$

Correct tension from correct working

A1

3

- (b) No change, as the ratios of the distances from the pivot would be the same

No

Reason

B1 B1

2

[8]

Q35.

(a) $R + 100 \sin 20^\circ + 60 \sin 50^\circ = 490$

Resolving vertically

Correct equation

M1 A1

$$R = 410$$

A1

3

(b) (i) $100 \cos 20^\circ - 60 \cos 50^\circ - F = 25$

Four term equation of motion

Correct equation

M1 A1

$$F = 30.4$$

Solving for F

Correct F from from correct working

M1 A1

4

(ii) $30.40 = \mu \times 409.8$

Use of $F = \mu R$

M1

$$\mu = 0.0742$$

Correct μ

A1



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